

17. Implement Mobile Price Prediction using appropriate machine learning algorithm

Aim

To implement a **Mobile Price Prediction model** using a suitable machine learning algorithm.

Software Required

- Python 3.x
- Pandas
- Scikit-learn

Procedure

1. Create a mobile phone dataset.
2. Use features such as RAM, storage and battery.
3. Define the price range as the target.
4. Split the dataset into training and testing sets.
5. Train a classification model.
6. Predict the mobile price range.
7. Calculate accuracy.

Code

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

data = {
    'RAM': [2, 3, 4, 6, 8, 12, 4, 6, 8, 12],
    'Storage': [32, 64, 64, 128, 128,
                256, 64, 128, 256, 256],
    'Battery': [3000, 3500, 4000, 4500, 5000,
                5000, 4000, 4500, 5000, 6000],
    'PriceRange': ['Low', 'Low', 'Medium', 'Medium',
                   'High', 'High', 'Medium', 'High',
                   'High', 'High']
```

```

}
df = pd.DataFrame(data)
X = df[['RAM', 'Storage', 'Battery']]
y = df['PriceRange']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Price Ranges:")
print(y_pred)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
sample = [[8, 256, 5000]]
print("\nPredicted Price Range:",
      model.predict(sample)[0])

```

Output:

```

Predicted Price Ranges:
['High' 'Low' 'High']

```

```

Accuracy: 1.0

```

```

Predicted Price Range: High

```

Result

Thus, the **Mobile Price Prediction model** was successfully implemented using a Decision Tree classifier.